Maximum value of the least upper bound $U_{\text{lub}}(A)$

Theorem

If the processor utilization factor of a task set Γ is larger than $1 \Rightarrow \Gamma$ is not feasible

Proof.

$U > 1 \Rightarrow UH > H$ since $H > 0 \Rightarrow \sum_{i=1}^{n} \frac{C_i}{T_i} H > H$ by U definition $\Rightarrow \sum_{i=1}^{n} \frac{H}{T_i} C_i > H$

$\frac{H}{T_i}$ is the (integer) number of times task τ_i is executed in the hyper-period

$\Rightarrow \frac{H}{T_i} C_i$ is the computation time requested by task τ_i in the hyper-period

$\Rightarrow \sum_{i=1}^{n} \frac{H}{T_i} C_i$ is the computation time requested by the task set in the hyper-period

If demand exceeds the available processor time H, the task set is not feasible $\square$

Remark

This result holds for any scheduling algorithm